Project ESCAPE: Does Entropy-Based Patching Find Syntax? Final Results Across Code and Natural Language

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Abstract

We test whether the entropy-triggered patch boundaries of the Byte Latent Transformer (BLT) align with syntactic constituent starts in Python, C++ and English, beyond a density-matched null and structure-blind baselines. On a frozen held-out split of 13,040 files, start F1 exceeds the null in every language (margins +0.120, +0.129, +0.348; $p \leq 10^{-4}$), but BLT underperforms it in all three languages against the strongest baseline: word-start splitting beats it on precision and recall at once, and indentation does better still in Python. Four further predictions fail as stated, and follow-up tests explain each. Precision is far below recall because the patcher proposes 2.6–9.8 boundaries per target. Prose aligns best (P3) because English constituents begin at words. The forced-opener reversal (P4) comes from scoring the wrong byte. The memory-unsafe residual (O5) exists only after conditioning on identifier entropy. A 14.3M-parameter student distilled from the 99.5M patcher keeps about 90% of the teacher's margin. Entropy boundaries are a genuine but shallow signal that follows word and line onsets more than syntax.

1 Introduction

Byte-level models remove a fixed subword vocabulary, but they must still decide where to spend computation. BLT groups bytes into variable-length patches whenever a small next-byte model's entropy exceeds a threshold τ (Pagnoni et al., 2025). If those boundaries fall where syntactic constituents open, the allocation tracks structure. But rare identifiers, word onsets and formatting also raise entropy, so proximity alone cannot separate these explanations.

ESCAPE asks five questions from our proposal. O1/P1: do boundaries align with constituent starts beyond a null that keeps each file's own patch lengths? P2: more at starts than at ends? O2/P3: is code segmented differently from prose? P4: is alignment weaker for constructs whose opener is forced (`if`, `for`) than for open-ended ones? O3: does Python (indentation) differ from C++ (brackets)? O4: does alignment survive code noise? O5: does extra compute near memory-unsafe C++ survive conditioning on identifier unpredictability? This report completes O3–O5, explains the failed predictions with targeted tests, checks the statistics against clustering, parser-recovery and context artefacts, and, beyond the proposal, distils the patcher into a 7× smaller student evaluated under the same frozen protocol (§5).

Our contributions are: (i) a byte-exact extraction pipeline that restores the 512-byte sliding window missing from the public conversion; (ii) a calibration-first protocol with a density-matched null and structure-blind baselines as the burden of proof; (iii) held-out results in three languages, every number regenerated from retained artefacts; and (iv) follow-ups showing each failed prediction is a statement about which bytes carry entropy.

2 Background and Related Work

BLT segments bytes dynamically so that expensive latent computation operates over patches (Pagnoni et al., 2025). BPE fixes its merge table in advance (Sennrich et al., 2016); ByT5 and CANINE remove the vocabulary but process every byte uniformly (Xue et al., 2022; Clark et al., 2022). Dynamic token pooling learned boundaries from conditional-entropy spikes in small models (Nawrot et al., 2023); BLT applies an entropy threshold at scale, so each boundary is a claim that can be checked against external structure. Structural probes recover syntax from trained representations (Hewitt and Manning, 2019; Feng et al., 2020), but need a trained probe. We ask whether an unsupervised

compression signal with no parser in the loop recovers constituents as a side effect, as statistical word segmentation placed boundaries by predictability and was checked against observable word boundaries (Brent, 1999; Goldwater et al., 2009). We use tree-sitter ASTs for code and benepar parses (Kitaev and Klein, 2018) for English, and, for the extension, online knowledge distillation (Hinton et al., 2015).

3 Protocol

Data and checkpoint. The frozen manifest has 16,300 samples: 8,000 Python functions from CodeSearchNet (Husain et al., 2019), 8,000 C++ files from The Stack (Kocetkov et al., 2022) (the proposal's CodeSearchNet has no C++) and 300 WikiText-103 paragraphs (Merity et al., 2017), all restored and hash-verified. We load the 99.5M-parameter patcher of `itazap/blt-1b-hf` (revision `91aa6b8e`) in float32 rather than the proposal's bfloat16 (bf16 weights upcast losslessly) with $\tau = 1.3354$ nats, no length cap, and the reference 512-byte sliding window in 8,192-token chunks. Six of eight fidelity gates passed as declared, two criteria were amended, and the amended set passed on the cluster stack. The logits agree with an independent re-implementation to 1.05×10^{-4}, with identical boundaries.

Metric, null and baselines. Entropy at byte i is $H_i = -\sum_v p_\theta(v \mid x_{<i}) \log p_\theta(v \mid x_{<i})$ over 260 symbols. Predictions are interior entropy boundaries; with no length cap, every one after the forced first byte is entropy-triggered. Targets are the unique starts of non-wrapper constituents (outermost wins ties), excluding byte 0 and file end: statement- and expression-level tree-sitter nodes in code and benepar phrases in English. Parser offsets are round-tripped to UTF-8 byte offsets and validated. Memory-unsafe regions for O5 are C++ statements containing pointer dereference or address-of (`*`, `&`), `new`, `delete`, unsafe C library calls, or unsafe casts. Maximum-cardinality one-to-one matching within k bytes gives pooled P, R and F1. Each language is split 20/80 by seeded SHA256 rank into calibration and test files, and only calibration files choose $k \in \{0, 1, 2, 4, 8\}$, by the median per-file F1 margin over the null; $k = 0$ was selected everywhere. The null shuffles each file's (length, reason) patch pairs except the last, which stays fixed so that file end is never a boundary, and so it preserves the patch-length multiset and boundary count. We use 10,000 seeded permutations per file and 2,000 paired file-bootstrap replicates. A structural reading must also beat boundaries at every newline, at indentation changes, and at every word start under the same targets and matching. We call this frozen protocol C-v1; the frozen teacher run was scored once and no primary number changed after the mid-submission. Later test-split scorings are either pre-registered extension runs (§5), each scored once, or sensitivity and exploratory runs ($k = 1, 2$, the O4 rate sweep, HF full attention, the follow-up arms, a τ grid) that selected no setting.

4 Results on the Proposal's Objectives

P1 and the structure-blind burden. Start F1 exceeds the null in every language (Table 1), with ratios of 4–5× in code. Against the strongest structure-blind baseline, however, BLT underperforms it in all three languages (Table 2): Python -0.242 [-0.247, -0.237] against indentation, C++ -0.173 against word starts, and English -0.274 against word starts. At BLT's density the F1 comparison is partly settled before alignment enters: with r predictions per target, F1 $\leq 2/(r+1)$, which caps BLT at 0.184, 0.256 and 0.554, below each strongest baseline. A density-free comparison agrees: word-start splitting places fewer boundaries than BLT yet beats it on precision *and* recall in every language (Python 0.142/0.892 against 0.083/0.822). Span IoU, the proposal's secondary metric, agrees (Appendix Table 11).

Precision versus recall. The proposal's mechanism predicted precision above recall: cheap interiors and forced closing delimiters would rarely trigger, so boundaries would be few and well placed. Instead precision is below recall in all three languages (Python 0.083/0.822; C++ 0.095/0.646; English 0.342/0.894). Since $H \leq \log B$, every boundary sits where at least $e^\tau \approx 3.8$ continuations were live. Identifier interiors, strings and comments are flat without being syntactic, so predictions outnumber targets 9.8 to one in Python, 6.8 in C++ and 2.6 in English, which caps precision while the same density guarantees high recall. P2 holds in the predicted direction: end margins are negative (-0.017, -0.035, -0.079), so closing delimiters are anti-aligned.

Why P3 came out reversed. The proposal predicted large BPP variance in code and near-chance

Language	k	Test files	P	R	F1	H_0 F1	ΔF1 [95% CI]	p
Python	0	6,400	0.083	0.822	0.152	0.032	+0.120 [0.117, 0.122]	≤1e-4
C++	0	6,400	0.095	0.646	0.165	0.036	+0.129 [0.125, 0.134]	≤1e-4
English	0	240	0.342	0.894	0.495	0.147	+0.348 [0.341, 0.354]	≤1e-4

Table 1: Held-out entropy-boundary alignment with constituent starts (P1). k was frozen on the calibration split. H_0 F1 is the permutation mean of the fixed-EOF density-matched null (10,000 permutations); intervals are paired file bootstraps; p is the one-sided permutation value.

Lang.	BLT	NL	Indent	Word	BLT−best [95% CI]
Python	0.152	0.000	0.394	0.245	-0.242 [-0.247, -0.237]
C++	0.165	0.041	0.281	0.339	-0.173 [-0.181, -0.167]
English	0.495	0.000	0.008	0.769	-0.274 [-0.280, -0.268]

Table 2: Start F1 of structure-blind baselines under the same targets, k and matching: boundary after every newline (NL), at indentation changes, and at every word start. The last column compares BLT with the strongest baseline for that language. Newline scores 0 in Python because the byte after a newline is indentation, never a start at $k = 0$; English paragraphs contain no newline.

alignment in prose. Mean bytes per patch (BPP) is 7.86 in C++, 5.68 in Python and 3.77 in English. The first half holds: within-file BPP variance is 527.8 in C++ and 69.4 in Python against 10.3 in English, because boilerplate stretches alternate with dense logic. The second half fails: the prose margin +0.348 exceeds the code margin +0.129. Two facts explain this. First, a benepar constituent always begins at a word, and the patcher fires at word onsets, so the word-start rule alone reaches F1 0.769 in English. Second, normalising does not rescue P3. Because the null keeps BLT's boundary count, F1 over null F1 equals R/R_0, which cannot exceed $1/R_0$ (5.8, 7.1, 3.8), close to the mean patch length. The raw ratio favours code (4.7× Python, 4.6× C++, 3.4× English) only because code patches are longer. Chance-corrected recall $(R - R_0)/(1 - R_0)$ has no such ceiling and ranks English first (0.856 against 0.784 and 0.588). P3 is reversed: prose aligns best, and it does so at word onsets.

Example and deliverable (4). The patcher's segmentation of a short Python function, with patch starts marked by `|`:

```
d|e|f |f|i|b|onacci|(|n|):
|if n |< |2:
|return |n|
|return |fibonacci(n |- 1) |+
|fibonacci(n - 2)
```

Boundaries open statements after indentation, and the bytes after `if` and `return` stay inside one patch. An identifier is split at nearly every byte on first use and kept whole on reuse. Figure 1 shows per-byte entropy heatmaps for a Python file, a C++ file and an English paragraph; BPP and patch-length distributions are in Appendix Figure 6.

Why P4 came out reversed, and what the forced bytes do. By type, forced openers have larger start-recall margins than open-ended constructs (Appendix Table 7; mean +0.759 vs. +0.611 in Python, +0.691 vs. +0.451 in C++), the opposite of P4. The mid-report noted that a start boundary sits *before* the keyword, where the choice of statement is still open. The forcing the proposal predicted applies to the bytes after the keyword, so we tested those directly. For every primary target start s we take its first token (the identifier run at s: the keyword for forced openers) and score the boundary rate at fixed positions against the same 10,000 null draws as the frozen scorer, so the start rows reproduce the frozen type recalls exactly (Appendix Table 6). The rest of the keyword is segmented below chance (Python -0.130, C++ -0.088), and the byte right after it almost never starts a patch (rate 0.0002 and 0.035 against null rates of 0.159 and 0.140). The boundary reappears where choice reopens. In Python the next non-space byte begins the condition, and its boundary rate is 0.895 (+0.736). In C++ the next byte is the equally forced `(`, and the boundary moves to the condition at offset four (rate 0.377; Figure 2). P4's mechanism therefore holds at the bytes it is about: once a keyword is committed, its continuation is predictable and is segmented far below chance. A start-offset metric cannot see this: the start byte is where forcing has not yet happened.

O3: Python versus C++ on a common depth scale. Raw tree-sitter depth is not comparable across languages, because C++ statements sit under more scaffolding nodes. We therefore count depth in constituents, as the number of stored target-type nodes that strictly enclose a node, and also

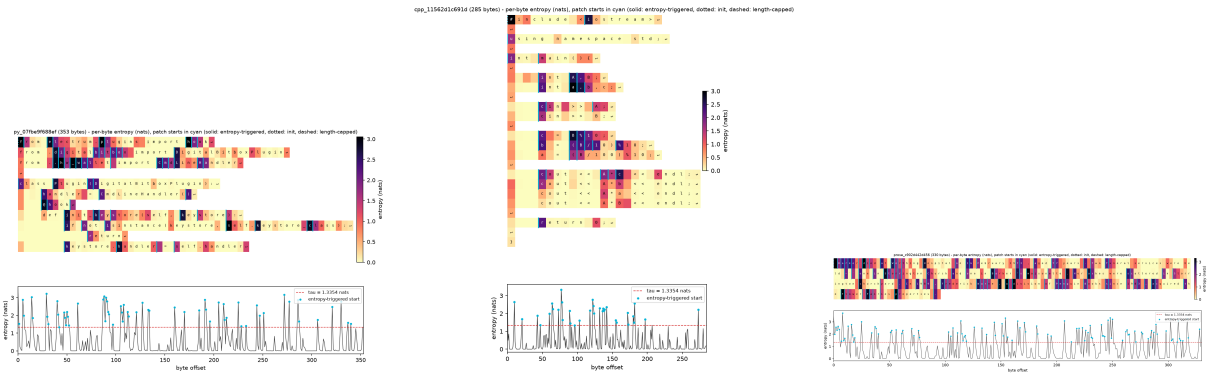


Figure 1: Per-byte entropy heatmaps for a Python file, a C++ file and an English paragraph (frozen teacher run).

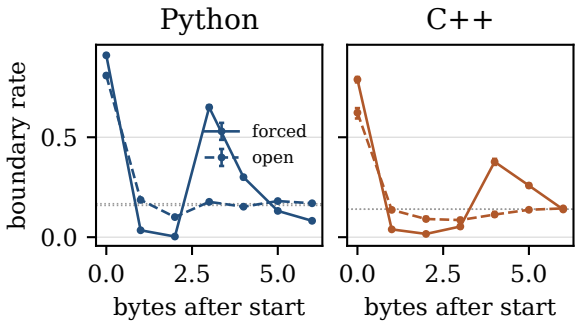


Figure 2: Boundary rate by byte offset after a constituent start: forced openers (solid), open-ended (dashed), null (dotted).

normalise it within each file (Figure 3). On this common scale all three languages align most at the outermost constituents. From one enclosing constituent to seven, the margin falls from +0.707 to +0.519 in Python, from +0.553 to +0.387 in C++, and from +0.714 to +0.601 in English; C++ falls further, to +0.151, at eight or more. Python exceeds C++ at every depth both languages share and at every relative-depth decile. English is nearly flat below the top levels. The indentation-versus-brackets contrast that O3 asked about shows up in the baselines rather than in the depth profile. In Python the indentation rule is the strongest baseline, because indentation *is* the grammar. In C++ word starts are the strongest baseline, and BLT's deficit to them (-0.173) is smaller than its deficit to indentation in Python (-0.242).

O4: robustness to noise, by rate. At the frozen rates (20% of lines or keywords, 50% of identifiers) every noised margin stays at or above +0.099 (Python) and +0.118 (C++), and the largest drop is -0.016 for indentation noise in Python (Appendix Table 10). The perturbations are the proposal's: missing semicolons (C++), indentation changes, camelCase↔snake_case inversion, and keyword typos. To test whether this "graceful degradation" depends on the rate, we rebuilt the same 1,000 files per language with each noise type at 5, 10, 20 and 40% (case inversion at 25, 50 and 100%) and rescored them with the unchanged O4 scorer (Figure 4). Indentation noise does by far the most damage, growing with the share of lines touched: from -0.005 at 5% to -0.029 at 40% in Python and from -0.004 to -0.023 in C++, the signature of a patcher whose statement boundaries ride on line layout. At 40%, keyword typos cost -0.007 (Python) and -0.002 (C++), missing semicolons -0.003, and renaming at most -0.001 and -0.002. Typos also lower bytes per patch (-0.43, -0.60 at 40%). Every noised margin stays at or above +0.086.

O5: is there a semantic residual near memory-unsafe code? We first audited the region labels (`docs/O5_LABEL_AUDIT.md`): re-running the queries reproduced 12,609/12,609 frozen regions exactly, and an independent lexical labeller agreed on 433/435 audited statements. Following the proposal's compute-profiling metric we model two outcomes, boundary density and inverse-BPP compute density. The frozen regression gives unsafe statements +0.0069 [0.0043, 0.0095] extra boundaries per byte after file fixed effects and identifier controls. Refitting it as a nested sequence (our refit reproduces the frozen coefficients) changes the reading (Appendix Table 8). With file fixed effects only, unsafe statements receive *fewer* boundaries (-0.0083 [-0.0127, -0.0039]), because they use more predictable identifiers. The positive coefficient appears only after conditioning on identifier entropy, and identifier share alone does not flip it. The conditional effect is 4.2% of the comparison statements' mean density. On inverse BPP, the proposal's compute metric, the file-fixed-effects estimate is null

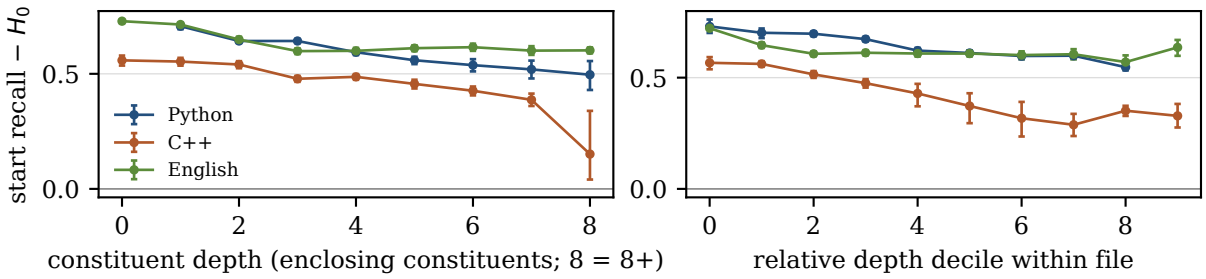


Figure 3: O3: held-out start recall minus mean H_0 recall by scaffold-free depth. Left: number of enclosing constituents (raw tree-sitter depth also counts grammar scaffolding that differs by language). Right: within-file relative depth decile. Strata with at least 200 starts; file-bootstrap 95% intervals.

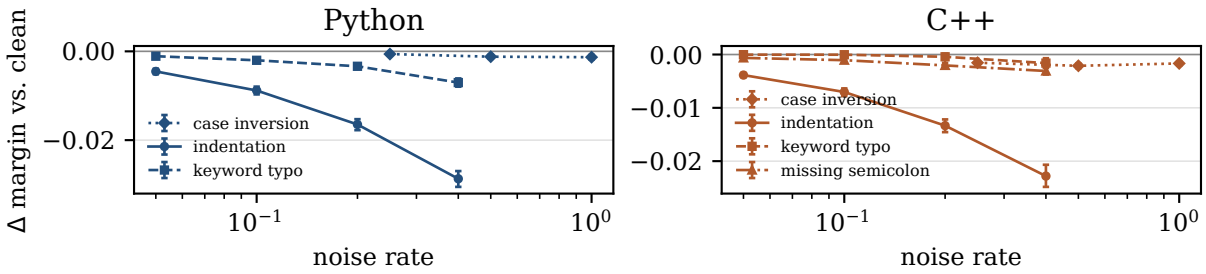


Figure 4: O4 rate sweep: change in start margin over H_0 relative to clean files, by noise type and rate (log scale), on 1,000 test files per language; paired bootstrap 95% intervals.

(-0.0024 [-0.0066, 0.0017]). O5 therefore finds no evidence that the patcher allocates extra compute to memory risk: what survives the confound is a small conditional effect, and the unconditional effect is negative or null.

Are the intervals too narrow? The 6,400 Python test functions come from only 315 repositories. Resampling repositories widens the Python interval 3.7×, from [0.118, 0.122] to [0.112, 0.127], which still excludes zero. Tree-sitter error recovery affects 28.0% of C++ test files; their margin differs from clean parses by +0.004 [-0.005, 0.015]. Files over 8,191 bytes, whose context resets at chunk edges, are 31.5% of C++ test files but 86% of its bytes, and their margin differs by -0.002 [-0.008, 0.004]. Under $k = 1, 2$ the margins shrink and BLT never overtakes the strongest baseline. HF's native full-attention mode, not faithful beyond 512 bytes, cuts the margin to +0.023 (Python), +0.008 (C++) and +0.150 (English), against +0.120, +0.131 and +0.348 with the restored window, so every result depends on that restoration.

5 Beyond the Proposal: a Distilled Patcher

The proposal needed no training. As a separately pre-registered extension, we ask whether a much smaller byte model can reproduce the patcher's boundaries and whether structural alignment survives distillation. A 14.3M-parameter student (384 dimensions, 8 layers, 6 heads, BLT-style byte transformer) was trained from scratch by online distillation, $\alpha\,\mathrm{CE} + (1-\alpha)T^2\,\mathrm{KL}(p_T \| p_S)$ with $\alpha = 0.1$ and $T = 2$, on ≤512-byte windows of a disjoint extension corpus. That corpus has 29,000 files: CodeSearchNet functions from repositories not in the evaluation set, WikiText paragraphs from unused articles, and Stack C++ from other repositories, split by repository or article. On held-out extension windows the student's boundaries reach F1 0.871 against the teacher's. Dropping the teacher ($\alpha = 1$, same data, seed and schedule) reaches 0.789, and placing a boundary at every word start reaches 0.475. The $\alpha = 1$ student predicts the true next byte better (1.17 vs. 1.24 bits/byte; teacher 1.14), so distillation trades a little compression for agreement with the teacher's segmentation.

Held-out alignment. We extracted the student on all 16,300 corpus files with stride-1 fresh 512-byte windows, the regime it was trained in. The extractor first reproduced the committed golden-fixture scores on the cluster GPU (largest F1 deviation 0.0000). The student then went through the frozen C-v1 pipeline with its own calibration

	Teacher (BLT patcher, 99.5M)				Student (distilled, 14.3M)					
Lang.	P	R	F1	Δ	P	R	F1	Δ	Student−teacher Δ [95% CI]	vs. best base.
Python	0.083	0.822	0.152	+0.120	0.076	0.859	0.140	+0.109	-0.011 [-0.011, -0.010]	-0.253
C++	0.095	0.646	0.165	+0.129	0.087	0.732	0.155	+0.119	-0.011 [-0.012, -0.009]	-0.183
English	0.342	0.894	0.495	+0.348	0.301	0.965	0.459	+0.304	-0.044 [-0.048, -0.040]	-0.310

Table 3: The distilled student on the frozen held-out split, scored by the frozen C-v1 pipeline with its own calibration record (which also selected $k = 0$ in every language). Δ is F1 minus the mean H_0 F1 (10,000 fixed-EOF permutations); the student−teacher interval is a paired file bootstrap; the last column is the student's F1 minus the strongest structure-blind baseline (indentation for Python, word-start otherwise).

Model	Lang.	τ^*	k^*	F1 at τ=1.335	F1 at τ^*	Δ at τ^* [95% CI]	vs. best baseline [95% CI]
Teacher	Python	2	0	0.152	0.184	+0.153 [0.150, 0.155]	-0.210 [-0.215, -0.205]
Teacher	C++	2	0	0.165	0.190	+0.158 [0.153, 0.163]	-0.148 [-0.156, -0.142]
Teacher	English	2	0	0.495	0.554	+0.426 [0.419, 0.433]	-0.215 [-0.221, -0.210]
Student	Python	2	0	0.140	0.178	+0.147 [0.144, 0.150]	-0.216 [-0.221, -0.211]
Student	C++	2	0	0.155	0.190	+0.156 [0.151, 0.160]	-0.149 [-0.155, -0.143]
Student	English	2.5	0	0.459	0.583	+0.460 [0.453, 0.468]	-0.186 [-0.193, -0.180]

Table 4: The single confirmatory test-split run at the pre-registered τ^* (argmax over $\{0.8, 1.0, 1.335, 1.7, 2.0, 2.5\}$ of the median per-file margin on calibration files only, ties to the smaller τ), with k re-frozen on the calibration split under τ^* by the frozen criterion. This run was scored once per model.

Model	F1	H_0 F1	Δ	vs. word
Teacher	0.500	0.150	+0.350	-0.273
Student	0.463	0.158	+0.305	-0.311

Table 5: The confirmatory English result: the pre-registered holdout of 4,107 paragraphs from 120 WikiText-103 articles unused by evaluation and distillation, scored once at the frozen $k = 0$ (10,000 permutations; all $p \leq 10^{-4}$). Word-start F1 = 0.774; the last column is F1 minus word-start F1.

record, which again selected $k = 0$ everywhere (Table 3). It keeps 91%, 92% and 87% of the teacher's margin (Python, C++, English). The paired deficit is small but resolved (-0.011 [-0.011, -0.010] in Python), and the student, like the teacher, stays below the baselines. A control on 300 test files per code language restricts the *teacher* to the student's fresh 512-byte windows. This costs +0.004 [0.003, 0.005] (Python) and +0.004 [0.003, 0.005] (C++) of F1, about 38% and 43% of the student's deficit. The student's forward pass is 5.0× faster than the teacher's on one RTX 2080 Ti.

Threshold and tolerance. The checkpoint's τ was set for compression, not alignment. We swept $\tau \in \{0.8, 1.0, 1.335, 1.7, 2.0, 2.5\}$ on calibration files only; τ^* was pre-registered as the argmax of the median per-file margin, ties to the smaller τ. For the teacher $\tau^* = 2$ in all three languages (calibration gain +0.037 in Python), and for the student $\tau^* = 2$ (Python), 2 (C++) and 2.5 (English; Figure 5). The student's English optimum sits at the edge of the grid, where its entropy runs higher than the teacher's. The grid was not extended, and each language was confirmed at its own τ^*. After re-freezing k under τ^*, each model's confirmatory test run was scored exactly once (Table 4): the teacher's margin rises to +0.153 (Python), +0.158 (C++) and +0.426 (English), a paired gain of +0.033 [0.032, 0.034] in Python, and it remains below the strongest baseline (-0.210, -0.148, -0.215). The student's margin at its own τ^* is +0.147, +0.156 and +0.460, also below the baselines. A later test-file run of the whole grid, which selects nothing, peaks at the same τ^* . Fewer, higher-entropy boundaries raise precision and margin, but no threshold makes entropy a parser.

The confirmatory English test. Because the 300 evaluation paragraphs were inspected during exploration, we pre-registered a holdout of the 120 lowest-ranked WikiText articles that neither evaluation nor distillation used, 4,107 paragraphs with benepar structure, and scored it once after team sign-off (Table 5). The teacher's margin is +0.350, against +0.348 on the evaluation paragraphs, and the student's is +0.305. Both stay below word starts (-0.273 and -0.311), so the English findings replicate on untouched text.

Byte balancing and domain adaptation. Two exploratory follow-ups ran on the 600-file code subset of the earlier control plus the English

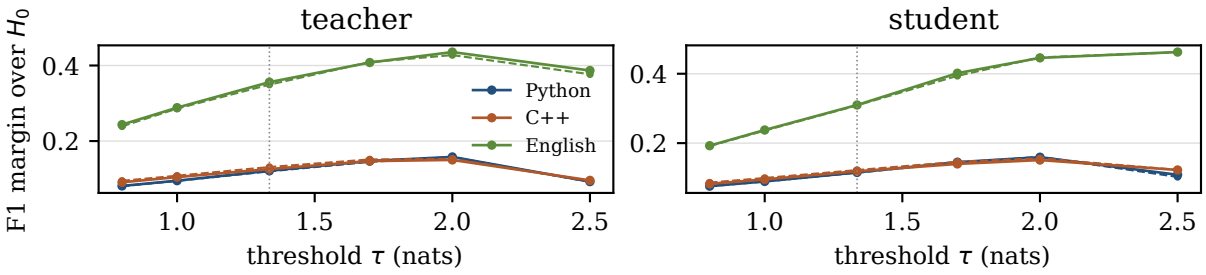


Figure 5: Threshold sweep: median per-file F1 margin over H_0. Solid: calibration files, which fixed τ^*. Dashed: the same grid on test files (exploratory); every point trails the strongest baseline by at least 0.148 F1. Dotted: $\tau = 1.335$.

test split. C++ is 73% of the distillation bytes. A student trained with every domain drawn to an equal expected byte share agrees with the teacher on held-out windows at F1 0.860 (distilled student: 0.871); its margin differs from the distilled student's by -0.002 [-0.003, -0.001] (Python), -0.002 [-0.0035, -0.0001] (C++) and +0.010 [0.008, 0.012] (English), so balancing trades code for prose. The pre-registered adaptation arm initialises the teacher architecture from the patcher and trains it with the same loss and data; its margin differs from the student's by +0.006 [0.004, 0.008], +0.005 [0.003, 0.007] and +0.029 [0.026, 0.032]. The criterion, a gain larger than the width of the student's confidence interval in at least two domains, is not met (1 of 3): in code, fine-tuning only recovers the unadapted teacher read through the same windows.

6 Verdict on H1 and Deliverables

The proposal defined support for H1 as: P1 clears both baselines with non-overlapping intervals, P2 holds in the predicted direction, P3 shows a code/prose dissociation, and P4's stratification appears. Judged that way, H1 is not supported: only P2 holds. P1 clears the null (H0) but not the whitespace and word-start baselines; P3 and P4 are reversed; and precision is below recall. All five proposal deliverables are delivered: (1) the extraction pipeline (`stream_a`); (2) the tree-sitter and benepar mapping engine with the memory-unsafe queries (`stream_b`); (3) the evaluation harness (`escape_scoring`, `escape_eval`); (4) the entropy heatmaps and BPP profiles; and (5) this report and the presentation, covering O1–O5. Code, per-file results, tests and the build that regenerates every number ship in `Trimax-Final.zip`; the three distilled students are on Hugging Face (`https://huggingface.co/Devatri/escape-blt-entropy-student`; with the suffixes `-alpha1` and `-balanced`, the α=1 control and the byte-balanced student), with checkpoints and training curves in the W&B project `escape-blt-patching`.[1]

7 Conclusion

BLT's entropy boundaries sit at constituent starts 3.4 to 4.7 times more often than a null with the same patch lengths, in all three languages, on held-out files and on a pre-registered English holdout, and a 7× smaller distilled student keeps about 90% of that margin. Yet indentation and word-start rules segment better at every threshold. The failed predictions have mechanical causes: several boundaries per target (precision), word onsets (P3), forcing that acts after the opener (P4), and a residual that appears only after conditioning on identifier entropy (O5). Entropy-based patching tracks uncertainty, which syntax shapes without being what it measures.

Ethical Considerations

We analyse existing text and code without executing corpus programs. Dataset provenance and licences accompany restored inputs.

Limitations and Future Work

Limitations. The checkpoint is a third-party conversion with a sliding window we restored; bitwise identity with the gated original was not tested, and its patch-length distribution was not compared with statistics published for the original model, as the proposal planned. English targets are predicted benepar parses, and the 300 evaluation paragraphs

[1] `https://forge.coreweave.com/wandb/kanhadivyansh123-iiit-hyderabad/escape-blt-patching`

were seen during exploration, so only the holdout is confirmatory. The O5 labels are syntactic (no types, aliasing or bounds) and restricted to clean-parse C++. The student was evaluated at one size, with the rolling extractor only. The $\alpha = 1$ control and threshold sweep share the corpus and seed, so between-seed variance is not measured. The follow-up arms used a 600-file code subset and the English test split. We did not evaluate a subword-tokenizer baseline, so O4 shows robustness only to the tested perturbations. Intervals are not corrected for multiple comparisons, so small effects such as the byte-balanced student's C++ difference may not survive. English intervals resample paragraphs, not articles; C++ intervals are not clustered by repository.

Future work. A parser-supervised teacher was not run. Open items are a subword-tokenizer baseline for O4, several student seeds, full-corpus scoring of the follow-up arms, a single-pass student kernel, and a bitwise check against the gated checkpoint.

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A Supplementary Tables

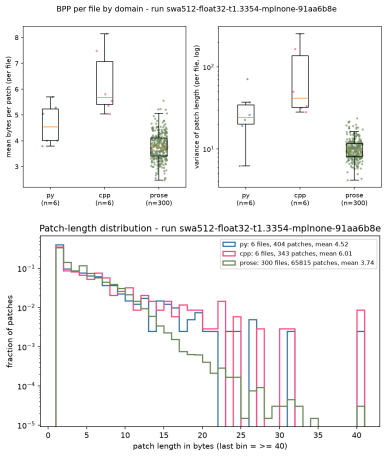


Figure 6: BPP per file and patch-length distributions per language (frozen teacher run).

Lang.	Position	Forced opener rate	Δ	Open-ended rate	Δ
Python	start s	0.910	+0.751	0.809	+0.633
	rest of first token	0.029	-0.130	0.094	-0.065
	byte after token	0.0002	-0.159	0.156	-0.014
	next non-space byte	0.895	+0.736	0.178	-0.001
C++	start s	0.789	+0.649	0.623	+0.483
	rest of first token	0.048	-0.088	0.087	-0.050
	byte after token	0.035	-0.105	0.139	-0.001
	next non-space byte	0.161	+0.021	0.241	+0.101

Table 6: P4 follow-up: boundary rate at fixed positions relative to each constituent start s (intervals are in the text), against the same 10,000 null draws as the frozen scorer (Δ = rate $-$ H_0 rate). The first token is the identifier run at s (the keyword for forced openers).

Lang.	Type	Opener	Starts	R	ΔR
Python	`assignment`	open	39	0.872	+0.715
Python	`binary_operator`	open	7,276	0.683	+0.498
Python	`call`	open	33,643	0.766	+0.598
Python	`expression_statement`	open	43,709	0.864	+0.695
Python	`for_statement`	det.	2,829	0.934	+0.772
Python	`function_definition`	det.	485	0.891	+0.724
Python	`if_statement`	det.	10,084	0.903	+0.745
Python	`return_statement`	open	7,010	0.798	+0.550
Python	`while_statement`	det.	263	0.954	+0.794
C++	`assignment_expression`	open	8,078	0.452	+0.319
C++	`binary_expression`	open	285,064	0.506	+0.362
C++	`call_expression`	open	456,491	0.595	+0.457
C++	`declaration`	open	249,479	0.773	+0.632
C++	`expression_statement`	open	486,954	0.655	+0.517
C++	`for_statement`	det.	21,326	0.823	+0.686
C++	`function_definition`	det.	101,613	0.768	+0.635
C++	`if_statement`	det.	126,003	0.795	+0.649
C++	`return_statement`	open	93,611	0.562	+0.418
C++	`while_statement`	det.	4,102	0.942	+0.792

Table 7: Start recall R by constituent type (P4); ΔR is recall minus the null mean. Types are rematched separately, so rows are not additive. A start shared by several nodes is credited to the outermost one (hence few Python `assignment` starts).

Model	Boundary density	Inverse BPP
file FE	-0.0083 [-0.0127, -0.0039]	-0.0024 [-0.0066, 0.0017]
+ stmt type	-0.0078 [-0.0122, -0.0035]	-0.0021 [-0.0062, 0.0020]
+ id. entropy	+0.0063 [0.0037, 0.0089]	+0.0109 [0.0084, 0.0133]
+ id. share	-0.0080 [-0.0123, -0.0036]	-0.0022 [-0.0063, 0.0019]
frozen	+0.0069 [0.0043, 0.0095]	+0.0112 [0.0087, 0.0137]

Table 8: O5: the memory-unsafe coefficient [95% CI] as controls are added (OLS, file fixed effects, file-clustered CR1 intervals; 12,609 statements in 1,500 clean-parse C++ test files, 5,271 unsafe). The last row is the frozen model.

Lang.	Files	Arm	F1	Δ
Python	300	Teacher (native)	0.143	+0.114
		Teacher, 512 windows	0.139	+0.110
		Student (distilled)	0.132	+0.103
		Student, byte-balanced	0.130	+0.101
		Teacher, adapted (512 windows)	0.139	+0.109
C++	300	Teacher (native)	0.162	+0.125
		Teacher, 512 windows	0.158	+0.121
		Student (distilled)	0.152	+0.115
		Student, byte-balanced	0.150	+0.114
		Teacher, adapted (512 windows)	0.158	+0.121
English	240	Teacher (native)	0.495	+0.348
		Student (distilled)	0.459	+0.304
		Student, byte-balanced	0.467	+0.314
		Teacher, adapted (512 windows)	0.482	+0.333

Table 9: Follow-up arms on the pre-registered subset (300 test files per code language, 600–2,000 bytes) and the English test split, $k = 0$, 10,000 permutations. All models except the native teacher see fresh 512-byte windows.

Lang.	Noise	Edits	ΔF1 vs. H_0	Change vs. clean [95% CI]
Python	clean	0	+0.115	–
Python	camel↔snake	4,601	+0.114	-0.001 [-0.002, -0.001]
Python	indentation	4,416	+0.099	-0.016 [-0.018, -0.015]
Python	keyword typo	1,626	+0.112	-0.003 [-0.004, -0.003]
C++	clean	0	+0.131	–
C++	camel↔snake	83,037	+0.129	-0.002 [-0.003, -0.002]
C++	indentation	68,746	+0.118	-0.013 [-0.015, -0.012]
C++	keyword typo	17,884	+0.131	-0.0004 [-0.0010, 0.0001]
C++	missing `;`	25,972	+0.129	-0.002 [-0.002, -0.002]

Table 10: Objective 4: robustness to code noise on 1,000 held-out test files per language (clean margin differs slightly from Table 1). Each noise type is applied alone; targets are clean constituents mapped to noised offsets. The null uses 1,000 permutations per file; intervals are paired bootstraps of noised minus clean.

Lang.	Files	BLT	H_0	Line	Word	BLT$-H_0$ [95% CI]
Python	6,400	0.351	0.342	0.546	0.342	+0.008 [0.007, 0.010]
C++	6,188	0.372	0.356	0.606	0.409	+0.016 [0.013, 0.018]
English	240	0.377	0.334	0.049	0.397	+0.042 [0.040, 0.045]

Table 11: Secondary metric: mean best single-patch IoU over unique constituent spans, held-out split (files with at least one span). H_0 averages 100 fixed-EOF permutations; Line and Word are newline and word-start segmentations.